

Tho Nguyen

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Experience

Smart Home Security Monitoring System

Davis, CA

Full Stack Developer

January 2026 - Present

- Constructed a full stack home safety system integrating 4 sensor types and camera live feeds, leveraging AWS Rekognition to perform real time hazard detection of unsafe conditions within 5 seconds
- Designed a cloud connect user facing application feeding live data into cloud data pipelines to ingest and process real time sensor streams via AWS services, delivering 85% uptime for remote monitoring
- Programmed an Android application in Kotlin using Android Studio and AWS Amplify, implementing a live camera feed, push alert system, account login, and remote monitoring dashboard
- Led a team using Jira to assign tasks, track progress, and manage production, maintaining a scheduled delivery across 9 weeks of production.

Intelligent Pathfinding & Game AI System

Davis, CA

Full Stack Developer

January 2026 - March 2026

- Trained a reinforcement learning agent for Pac-Man improving decision processing to model an adaptive behavior through iterative reward sequences, scoring above 50% of all contestant AIs
- Engineered an optimized A* based pathfinding system with configurable heuristics, improving traversal efficiency by 61%
- Designed a minimax algorithm with alpha-beta pruning to remove unnecessary branches and deepen search depth, producing a strong competitive AI engine capable of strategic actions

Personal Portfolio

San Jose, CA

Frontend Developer

January 2026 - Present

- Implemented and deployed a responsive portfolio website to showcase technical projects and skills using Figma, HTML, CSS, React.js and Tailwind CSS, creating a personal display for recruiters
- Organized and structured all project content with intuitive UI components, improving usability and presentation for public access

Computer Network Protocols

Davis, CA

Backend Developer

Sept 2025 - Dec 2025

- Built a custom UDP/TCP communication system from scratch with congestion control and error detection logic, reducing the packet loss moderately above the components benchmark by 10%
- Engineered a routing simulation engine to model and benchmark network performance across 2 and 4 algorithms allowing quantitative comparisons to evaluate trade offs within latency and throughput
- Developed a network diagnostic tooling using Wireshark to validate protocol behavior and identify bottlenecks

Animal Shelter Web Development

Davis, CA

Full Stack Developer

April 2024

- Architected and developed a full stack application using React, Node.js, HTML, and CSS within a 24 hour hackathon as part of a 4 person team
- Designed responsive UI components and implemented the backend logic for dynamic data handling, earning positive feedback from judges and staff on usability and features.

SKILLS

Programming Languages: C, C++, C#, Python, Javascript, Kotlin

Software & Systems: React, Node.js, Tailwind CSS, Wireshark, Android Studio

Concepts: Object Oriented Programming, Networking, Data Structure & Algorithm

Tools: Git, Docker, GitHub Actions, VS Code, Figma, Overleaf, Jira, AWS

EDUCATION

University of California Davis

Davis, CA

Bachelor of Science in Computer Engineering

June 2026

- **Relevant Coursework:** Operating System, Computer Networks, Artificial Intelligence, Computer Architecture, Data Structure, Algorithm Design and Analysis